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DEVICE, SYSTEM, AND METHOD

Abstract

A device includes: an appearance information acquisition unit configured to acquire measured appearance information on an appearance of a mobile body; a movement information acquisition unit configured to acquire movement information on movement of the mobile body; a determination unit configured to determine whether a deficiency has occurred in the measured appearance information; a complement unit configured to, in a case where the determination unit determines that the deficiency has occurred, use the movement information to complement the deficiency in the measured appearance information; and an estimation unit configured to estimate at least one of a position and an orientation of the mobile body by comparing reference appearance information on the appearance of the mobile body with the measured appearance information, and, in a case where the determination unit determines that the deficiency has occurred, compare the complemented measured appearance information with the reference appearance information.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-025047 filed on Feb. 22, 2024. The disclosure of the above-identified application, including the specification, drawings, and claims, is incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a device, a system, and a method for estimating at least one of a position and an orientation of a mobile body.

2. Description of Related Art

[0003] In a vehicle manufacturing process, a technique of causing a vehicle to travel via unmanned driving is known (for example, Japanese Unexamined Patent Application Publication No. 2017-538619 (JP 2017-538619 A)).

SUMMARY

[0004] When a mobile body, such as a vehicle, is moved by unmanned driving, processing of estimating a position or an orientation of the mobile body is executed. The position or the orientation of the mobile body can be estimated by comparing measurement information acquired by using a detector, such as a camera or a light detection and ranging (LiDAR), with reference information prepared in advance. However, in a case where a deficiency has occurred in the measurement information, when the measurement information is directly used to execute the estimation processing, there is a possibility that a correct estimation result cannot be obtained.

[0005] The present disclosure can be implemented as the following aspects.

[0006] (1) A first aspect of the present disclosure provides a device. The device includes an appearance information acquisition unit, a movement information acquisition unit, a determination unit, a complement unit, and an estimation unit. The appearance information acquisition unit is configured to acquire measured appearance information on an appearance of a mobile body from a detector configured to measure the appearance of the mobile body. The movement information acquisition unit is configured to acquire movement information on movement of the mobile body. The determination unit is configured to determine whether a deficiency has occurred in the measured appearance information. The complement unit is configured to, in a case where the determination unit determines that the deficiency has occurred, use the movement information to complement the deficiency in the measured appearance information. The estimation unit is configured to estimate at least one of a position and an orientation of the mobile body by comparing reference appearance information on the appearance of the mobile body with the measured appearance information. The estimation unit is configured to, in a case where the determination unit determines that the deficiency has occurred, compare the measured appearance information complemented by the complement unit with the reference appearance information.

[0007] With the device according to the aspect, the deficiency in the measured appearance

information can be complemented. Therefore, a situation can be suppressed in which a correct estimation result of the position or the orientation of the mobile body cannot be obtained due to the occurrence of the deficiency in the measured appearance information.

[0008] (2) In the device according to the above-described aspect, the determination unit may be configured to, in a case where the deficiency is detected from the measured appearance information, determine that the deficiency has occurred in the measured appearance information.

[0009] With the device according to the aspect, in a case where the deficiency is detected from the measured appearance information, the deficiency in the measured appearance information can be complemented.

[0010] (3) In the device according to the above-described aspect, the determination unit may be configured to, in a case where the determination unit predicts that the deficiency will occur in the measured appearance information, determine that the deficiency has occurred in the measured appearance information.

[0011] With the device according to the aspect, the deficiency in the measured appearance information can be complemented in a case where the occurrence of the deficiency in the measured appearance information is predicted.

[0012] (4) In the device according to the above-described aspect, the determination unit may be configured to, in at least one of a case where the mobile body is located at a predetermined place and a case where the mobile body passes through the predetermined place, predict that the deficiency will occur in the measured appearance information.

[0013] With the device according to the aspect, the prediction can be easily made that the deficiency will occur in the measured appearance information.

[0014] (5) In the device according to the above-described aspect, the predetermined place may be a place where a predetermined number of workers are present.

[0015] With the device according to the aspect, a situation can be suppressed in which a correct estimation result of the position or the orientation of the mobile body cannot be obtained in a place where a blind spot is likely to occur in the detector due to the worker.

[0016] (6) In the device according to the above-described aspect, the predetermined place may be a place where the worker gets on the mobile body to perform work.

[0017] With the device according to the aspect, a situation can be suppressed in which a correct estimation result of the position or the orientation of the mobile body cannot be obtained in a place where a blind spot is likely to occur in the detector due to the worker.

[0018] (7) The device according to the above-described aspect may further include a controller configured to, in a case where the complement of the deficiency in the measured appearance information is repeated a predetermined number of times, decelerate or stop the mobile body.

[0019] With the device according to the aspect, the decrease in the estimation accuracy can be suppressed by repeating the complement.

[0020] (8) In the device according to the above-described aspect, the controller may be configured to change the predetermined number of times in accordance with a movement state of the mobile body.

[0021] With the device according to the aspect, a situation can be suppressed in which the movement state of the mobile body is unstable.

[0022] (9) In the device according to the above-described aspect, the controller may be configured to change the predetermined number of times in accordance with magnitude of the deficiency.

[0023] With the device according to the aspect, the estimation accuracy is likely to be decreased as the magnitude of the deficiency in the measured appearance information is increased, and thus a significant decrease in the estimation accuracy can be suppressed by changing the predetermined number of times in accordance with the magnitude of the deficiency.

[0024] (10) In the device according to the above-described aspect, the controller may be configured to change the predetermined number of times in accordance with a distance between the detector

and the mobile body.

[0025] With the device according to the aspect, the estimation accuracy is likely to be decreased as the distance between the detector and the mobile body is increased, and thus a significant decrease in the estimation accuracy can be suppressed by changing the predetermined number of times in accordance with the distance between the detector and the mobile body.

[0026] (11) A second aspect of the present disclosure provides a system. The system includes a detector, an appearance information acquisition unit, a movement information acquisition unit, a determination unit, a complement unit, and an estimation unit. The detector is configured to measure an appearance of a mobile body. The appearance information acquisition unit is configured to acquire measured appearance information on the appearance of the mobile body from the detector. The movement information acquisition unit is configured to acquire movement information on movement of the mobile body. The determination unit is configured to determine whether a deficiency has occurred in the measured appearance information. The complement unit is configured to, in a case where the determination unit determines that the deficiency has occurred, use the movement information to complement the deficiency in the measured appearance information. The estimation unit is configured to estimate at least one of a position and an orientation of the mobile body by comparing reference appearance information on the appearance of the mobile body with the measured appearance information. The estimation unit is configured to, in a case where the determination unit determines that the deficiency has occurred, compare the measured appearance information complemented by the complement unit with the reference appearance information.

[0027] With the system according to the aspect, the deficiency in the measured appearance information can be complemented. Therefore, a situation can be suppressed in which a correct estimation result of the position or the orientation of the mobile body cannot be obtained due to the occurrence of the deficiency in the measured appearance information.

[0028] (12) A third aspect of the present disclosure provides a method. The method includes acquiring measured appearance information on an appearance of a mobile body by measuring the appearance of the mobile body, acquiring movement information on movement of the mobile body, determining whether a deficiency has occurred in the measured appearance information, complementing, in a case where a determination is made that the deficiency has occurred, the deficiency in the measured appearance information by using the movement information, estimating, in a case where a determination is made that the deficiency has occurred, at least one of a position and an orientation of the mobile body by comparing the complemented measured appearance information with reference appearance information on the appearance of the mobile body.

[0029] According to the method according to the aspect, the deficiency in the measured appearance information can be complemented. Therefore, a situation can be suppressed in which a correct estimation result of the position or the orientation of the mobile body cannot be obtained due to the occurrence of the deficiency in the measured appearance information.

[0030] The present disclosure can also be implemented in various aspects other than the device and the method. For example, the method can be implemented in an aspect of a system, a computer program, a recording medium on which a computer program is recorded, or the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0032] FIG. **1** is a diagram showing a configuration of a system according to a first embodiment;
[0033] FIG. **2** is a diagram showing a configuration of a vehicle according to the first embodiment;
[0034] FIG. **3** is a diagram showing a configuration of a server device according to the first embodiment;
[0035] FIG. **4** is a diagram showing a state in which a vehicle is moved by remote control in a factory;
[0036] FIG. **5** is a flowchart showing a processing procedure of traveling control of the vehicle according to the first embodiment;
[0037] FIG. **6** is a diagram showing a deficiency in measured appearance information;
[0038] FIG. **7** is a diagram showing contents of movement information;
[0039] FIG. **8** is a diagram showing a method of detecting the deficiency in the measured appearance information;
[0040] FIG. **9** is a diagram showing a method of complementing the deficiency in the measured appearance information;
[0041] FIG. **10** is a diagram showing a configuration of a system according to a second embodiment;
[0042] FIG. **11** is a diagram showing a configuration of a vehicle according to the second embodiment; and
[0043] FIG. **12** is a flowchart showing a processing procedure of traveling control of the vehicle according to the second embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

A. First Embodiment

[0044] FIG. **1** is a diagram showing a configuration of a system **10** according to a first embodiment. In the present embodiment, the system **10** is used in a factory that manufactures a mobile body and is used to move the mobile body via unmanned driving. In the present embodiment, the system **10** includes a vehicle **100**, a server device **200**, at least one external sensor **300**, and a process management device **400**. In the present embodiment, the vehicle **100** can be regarded as a “mobile body” according to the present disclosure. The server device **200** can be regarded as a “device” according to the present disclosure. The external sensor **300** can be regarded as a “detector” according to the present disclosure.

[0045] In the present disclosure, the “mobile body” means a movable body, for example, a vehicle or an electric vertical take-off and landing aircraft (so-called flying car). The vehicle may be a vehicle that travels using wheels or a vehicle that travels using a caterpillar, and is, for example, a passenger car, a truck, a bus, a two-wheeled vehicle, a four-wheeled vehicle, a tank, and a construction vehicle. The vehicle includes a battery electric vehicle (BEV), a gasoline vehicle, a hybrid electric vehicle, and a fuel cell electric vehicle. In a case where the mobile body is other than the vehicle, the expression of “vehicle” and “car” according to the present disclosure can be replaced with “mobile body” as appropriate, and the expression of “travel” can be replaced with “move” as appropriate.

[0046] In the present disclosure, the “unmanned driving” means driving that does not depend on a traveling operation performed by an occupant. The traveling operation means an operation related to at least any one of “traveling”, “turning”, and “stopping” of the vehicle **100**. The unmanned driving is implemented by automatic or manual remote control using a device located outside the vehicle **100** or by autonomous control of the vehicle **100**. The occupant who does not perform the traveling operation may get on the vehicle **100** that travels via the unmanned driving. Examples of the occupant who does not perform the traveling operation include a person who simply sits on a seat of the vehicle **100** and a person who performs work different from the traveling operation, such as assembly, inspection, or operation of switches, in a state of getting on the vehicle **100**. The driving via the traveling operation performed by the occupant may be referred to as “manned driving”.

[0047] In the present disclosure, the “remote control” includes “complete remote control” in which all the operations of the vehicle **100** are completely decided from the outside of the vehicle **100**, and “partial remote control” in which a part of the operations of the vehicle **100** is decided from the outside of the vehicle **100**. In addition, the “autonomous control” includes “complete autonomous control” in which the vehicle **100** autonomously controls the operation thereof without receiving any information from an external device of the vehicle **100**, and “partial autonomous control” in which the vehicle **100** autonomously controls the operation thereof by using the information received from the external device of the vehicle **100**.

[0048] FIG. 2 is a diagram showing a configuration of the vehicle **100**. In the present embodiment, the vehicle **100** is a battery electric vehicle and is configured to travel via the remote control. The vehicle **100** includes a vehicle control device **110** that controls the respective units of the vehicle **100**, an actuator group **120** that is driven under control of the vehicle control device **110**, a communication device **130** that communicates with the server device **200** via wireless communication, and an internal sensor group **140**.

[0049] The actuator group **120** includes at least one actuator. In the present embodiment, the actuator group **120** includes an actuator of a drive device that generates a propulsion force of the vehicle **100**, an actuator of a steering device that changes a traveling direction of the vehicle **100**, and an actuator of a brake device that generates a brake force of the vehicle **100**. In the present embodiment, the drive device includes a battery, a traveling motor driven by an electric power of the battery, and wheels rotated by the traveling motor. The actuator of the drive device includes the traveling motor.

[0050] The internal sensor group **140** includes at least one internal sensor. The internal sensor is a sensor mounted in the vehicle **100**. In the present embodiment, the internal sensor group **140** includes a vehicle speed sensor for detecting a speed of the vehicle **100** and a steering angle sensor for detecting a steering angle of the vehicle **100** as the internal sensors.

[0051] The vehicle control device **110** is configured by a computer including a processor **111**, a memory **112**, an input/output interface **113**, and an internal bus **114**. The processor **111**, the memory **112**, and the input/output interface **113** are connected to be bidirectionally communicable with each other via the internal bus **114**. The actuator group **120**, the communication device **130**, and the internal sensor group **140** are connected to the input/output interface **113**.

[0052] The processor **111** functions as a traveling controller **196** and a movement information transmission unit **197** by executing a computer program PG1 stored in advance in the memory **112**.

[0053] The traveling controller **196** controls the actuator group **120**. The traveling controller **196** can cause the vehicle **100** to travel by controlling the actuator group **120** in response to the operation performed by the occupant in a case where the occupant gets on the vehicle **100**. The traveling controller **196** can cause the vehicle **100** to travel by controlling the actuator group **120** using a traveling control signal received from the server device **200** regardless of whether the occupant gets on the vehicle **100**. In the present embodiment, the acceleration and the steering angle of the vehicle **100** are included in the traveling control signal as parameters. In another embodiment, the speed of the vehicle **100** may be included in the traveling control signal as the parameter instead of the acceleration of the vehicle **100** or in addition to the acceleration of the vehicle **100**.

[0054] The movement information transmission unit **197** acquires a measurement result from the internal sensor group **140** and transmits the measurement result of the internal sensor group **140** to the server device **200**. In the present embodiment, the movement information transmission unit **197** repeatedly executes, at a predetermined cycle, the acquisition of the measurement result and the transmission of the measurement result. The measurement result of the internal sensor group **140** includes the speed of the vehicle **100** measured by the vehicle speed sensor included in the internal sensor group **140**, the steering angle of the vehicle **100** measured by the steering angle sensor included in the internal sensor group **140**, and measurement times of the speed and the steering

angle.

[0055] FIG. 3 is a diagram showing a configuration of the server device **200**. The server device **200** is configured by a computer including a processor **201**, a memory **202**, an input/output interface **203**, and an internal bus **204**. The processor **201**, the memory **202**, and the input/output interface **203** are connected to be bidirectionally communicable with each via the internal bus **204**. The communication device **205** for communicating with the vehicle **100** via wireless communication is connected to the input/output interface **203**. In the present embodiment, the communication device **205** can further communicate with the external sensor **300** and the process management device **400** via wired communication or wireless communication.

[0056] The processor **201** functions as a measured appearance information acquisition unit **211**, a movement information acquisition unit **212**, a deficiency determination unit **213**, a complement unit **214**, an estimation unit **215**, and a remote controller **216** by executing a computer program PG2 stored in advance in the memory **202**. In the present embodiment, the remote controller **216** can be regarded as a “controller” according to the present disclosure.

[0057] The measured appearance information acquisition unit **211** acquires measured appearance information MG from the external sensor **300**. The measured appearance information MG is information on the appearance of the vehicle **100**, and is information obtained by measuring the vehicle **100**. In the present embodiment, the external sensor **300** is a LiDAR, and the measured appearance information is three-dimensional point cloud data.

[0058] The movement information acquisition unit **212** acquires the measurement result of the internal sensor group **140** from the vehicle **100**, and records the measurement result of the internal sensor group **140** in the memory **202**. In the present embodiment, the movement information acquisition unit **212** records, in time series, the measurement results of the internal sensor group **140** repeatedly transmitted from the vehicle **100** to the memory **202** at a predetermined cycle. In the following description, information on movement of the vehicle **100**, such as the speed or the steering angle of the vehicle **100**, will be referred to as movement information MV.

[0059] The deficiency determination unit **213** determines whether the deficiency has occurred in the measured appearance information MG acquired by the measured appearance information acquisition unit **211**. The occurrence of the deficiency in the measured appearance information MG means that a part of the appearance of the vehicle **100** represented by the measured appearance information MG is deficient.

[0060] The complement unit **214** uses the movement information MV to complement the deficiency in the measured appearance information MG in a case where the deficiency determination unit **213** determines that the deficiency has occurred in the measured appearance information MG. In the present disclosure, the complement of the deficiency in the measured appearance information MG includes the complement of all the deficiencies in the measured appearance information MG, as well as the complement of a part of the deficiencies in the measured appearance information MG. That is, in the present disclosure, the complement of the deficiency in the measured appearance information MG means complement of at least a part of the deficiencies in the measured appearance information MG.

[0061] The estimation unit **215** estimates at least one of a position and an orientation of the vehicle **100** by comparing the measured appearance information MG with reference appearance information RG stored in advance in the memory **202**. The reference appearance information RG is information on the appearance of the vehicle **100**. In the present embodiment, the reference appearance information RG is three-dimensional point cloud data. The reference appearance information RG can be generated by using, for example, CAD data representing the appearance of the vehicle **100**. The estimation unit **215** estimates both the position and the orientation of the vehicle **100**. In a case where the deficiency determination unit **213** does not determine that the deficiency has occurred in the measured appearance information MG, the estimation unit **215** compares the measured appearance information MG in which the deficiency is not complemented

by the complement unit **214** with the reference appearance information RG. In a case where the deficiency determination unit **213** determines that the deficiency has occurred in the measured appearance information MG, the estimation unit **215** compares the measured appearance information MG in which the deficiency is complemented by the complement unit **214** with the reference appearance information RG.

[0062] The remote controller **216** executes the remote control of the vehicle **100**. The remote controller **216** generates the traveling control signal for causing the vehicle **100** to travel via the remote control by using the estimation result of the estimation unit **215**. The remote controller **216** transmits the traveling control signal to the vehicle **100**.

[0063] As shown in FIG. **1**, the external sensor **300** is a sensor located outside the vehicle **100**. The external sensor **300** measures the appearance of the vehicle **100** and generates the measured appearance information including the appearance of the vehicle **100**. In the present embodiment, the external sensor **300** is a LiDAR, and the measured appearance information includes the three-dimensional point cloud data and a measurement time of the three-dimensional point cloud data. The external sensor **300** includes a communication device (not shown) and can communicate with the server device **200** via wired communication or wireless communication. The external sensor **300** transmits the measured appearance information to the server device **200** through the communication device. In the present embodiment, the external sensor **300** repeatedly executes, at a predetermined cycle, the generation of the measured appearance information and the transmission of the measured appearance information.

[0064] The process management device **400** executes management of an entire manufacturing process of the vehicle **100** in a factory FC. The process management device **400** is configured by at least one computer. The process management device **400** includes a database in which process information on the manufacturing process of the vehicle **100** is recorded. The process information includes, for example, identification information of the vehicle **100**, the contents of the manufacturing process, and information on a progress status of the manufacturing process. The process management device **400** includes a communication device (not shown) and can communicate with the server device **200** via wired communication or wireless communication.

[0065] FIG. **4** is a diagram showing a state in which the vehicle **100** is moved by the remote control in the factory FC. In the present embodiment, the remote control of the vehicle **100** is executed in the factory FC that manufactures the vehicle **100**. The factory FC includes a first place PL**1** and a second place PL**2**. The first place PL**1** is, for example, a place where the vehicle **100** is constructed, and the second place PL**2** is, for example, a place where the vehicle **100** is inspected. The vehicle **100** constructed in the first place PL**1** is in a state in which the vehicle **100** can travel via the remote control. The first place PL**1** and the second place PL**2** are connected by a track TR on which the vehicle **100** can travel. A plurality of external sensors **300** is installed in the vicinity of the track TR. The server device **200** can estimate the position and the orientation of the vehicle **100** by using the measurement result of the external sensor **300**. The position and the orientation of the vehicle **100** in the factory FC can be expressed by using coordinates of X, Y, and Z in a global coordinate system GC. The server device **200** can generate the traveling control signal for causing the vehicle **100** to travel based on the estimation result of the position and the orientation of the vehicle **100**, and transmit the traveling control signal to the vehicle **100**. The vehicle **100** can travel in response to the received traveling control signal. Therefore, with the system **10**, the vehicle **100** can be moved from the first place PL**1** to the second place PL**2** by the remote control without using a transport device, such as a crane or a conveyor. The vehicle **100** that has passed the inspection in the second place PL**2** is then shipped from the factory FC.

[0066] FIG. **5** is a flowchart showing a processing procedure of the traveling control of the vehicle **100**. FIG. **6** is a diagram showing a deficiency KS in the measured appearance information MG. FIG. **7** is a diagram showing the contents of the movement information MV. FIG. **8** is a diagram showing a method of detecting the deficiency KS in the measured appearance information MG.

FIG. 9 is a diagram showing a method of complementing the deficiency KS in the measured appearance information MG.

[0067] Steps **S110** to **S170** shown in FIG. 5 are repeatedly executed by the processor **201** of the server device **200**. Steps **S180** to **S190** are repeatedly executed by the processor **111** of the vehicle control device **110**. In step **S110**, the measured appearance information acquisition unit **211** acquires the measured appearance information MG from the external sensor **300**. In the present embodiment, the measured appearance information MG is the three-dimensional point cloud data. The measured appearance information MG includes a point cloud of the vehicle **100** as a control target and a point cloud other than the point cloud of the vehicle **100**. Examples of the point cloud other than the point cloud of the vehicle **100** include a point cloud of a road surface of the track TR, a point cloud of various facilities of the factory FC, and a point cloud of the worker of the factory FC.

[0068] In step **S120**, the deficiency determination unit **213** determines whether the deficiency KS has occurred in the measured appearance information MG. When a determination is made in step **S120** that the deficiency KS has occurred in the measured appearance information MG, the deficiency determination unit **213** advances the processing to step **S130**. When a determination is not made in step **S120** that the deficiency KS has occurred in the measured appearance information MG, the deficiency determination unit **213** skips step **S130** and advances the processing to step **S140**. As shown in FIG. 6, in a case where an obstacle OB, such as a person or an object, is present between the external sensor **300** and the vehicle **100**, a blind spot is generated in the external sensor **300** due to the obstacle OB, and the deficiency KS occurs in the measured appearance information MG. In the present embodiment, the deficiency determination unit **213** determines whether the deficiency KS has occurred in the measured appearance information MG by using at least one of the following determination methods **A1** to **A4**.

Determination Method **A1**

[0069] In the determination method **A1**, the deficiency determination unit **213** determines whether the deficiency KS has occurred in the measured appearance information MG by using the process information acquired from the process management device **400**. In the manufacturing process in which a predetermined number or more of workers are present in the vicinity of the vehicle **100**, there is a high possibility that the worker enters between the external sensor **300** and the vehicle **100**, and the blind spot is generated in the external sensor **300** due to the worker. Therefore, the deficiency determination unit **213** determines that the deficiency KS has occurred in the measured appearance information MG in a case where the current manufacturing process of the vehicle **100** is the manufacturing process in which a predetermined number or more of workers are present in the vicinity of the vehicle **100**. In the present embodiment, the process information includes information on a current manufacturing process of the vehicle **100** and information on the number of workers present in the vicinity of the vehicle **100** in the current manufacturing process. The deficiency determination unit **213** acquires the number of workers present in the vicinity of the vehicle **100** by using the process information acquired from the process management device **400**, and determines that the deficiency KS has occurred in the measured appearance information MG in a case where the number of workers present in the vicinity of the vehicle **100** is equal to or more than a predetermined number. The deficiency determination unit **213** may acquire, in advance, the process information including information on a place where the manufacturing process is executed on the vehicle **100** and information on the number of workers present in the vicinity of the vehicle **100** in the place where the manufacturing process is executed on the vehicle **100** from the process management device **400**, and determine that the deficiency KS has occurred in the measured appearance information MG in a case where the vehicle **100** passes through the place where the predetermined number or more of workers are present in the vicinity of the vehicle **100** and the manufacturing process is executed on the vehicle **100**.

Determination Method **A2**

[0070] In the determination method A2, the deficiency determination unit **213** determines whether the deficiency KS has occurred in the measured appearance information MG by using the process information acquired from the process management device **400**. In the manufacturing process in which the worker gets on the vehicle **100** to perform the work, there is a high possibility that the blind spot is generated in the external sensor **300** due to the worker who gets on the vehicle **100**. Therefore, the deficiency determination unit **213** determines that the deficiency KS has occurred in the measured appearance information MG in a case where the current manufacturing process of the vehicle **100** is the manufacturing process in which the worker gets on the vehicle **100** to perform the work. In the present embodiment, the process information includes information on the current manufacturing process of the vehicle **100** and information on whether the worker gets on the vehicle **100** to perform the work in the current manufacturing process. The deficiency determination unit **213** determines that the deficiency KS has occurred in the measured appearance information MG in a case where the current manufacturing process of the vehicle **100** indicated in the process information acquired from the process management device **400** is the manufacturing process in which the worker gets on the vehicle **100** to perform the work. The deficiency determination unit **213** may acquire, in advance, the process information including information on a place where a manufacturing process in which the worker gets on the vehicle **100** to perform the work is executed from the process management device **400**, and determine that the deficiency KS has occurred in the measured appearance information MG in a case where the vehicle **100** passes through the place where the manufacturing process in which the worker gets on the vehicle **100** to perform the work is executed.

Determination Method A3

[0071] In the determination method A3, the deficiency determination unit **213** determines that the deficiency KS has occurred in the measured appearance information MG in a case where the deficiency determination unit **213** predicts that the deficiency KS will occur in the measured appearance information MG. For example, a camera is installed at a place where each of the manufacturing processes of the vehicle **100** is executed, and the deficiency determination unit **213** acquires the number of workers present in the vicinity of the vehicle **100** in the current manufacturing process by analyzing an image acquired from the camera, and predicts that the deficiency KS will occur in the measured appearance information MG in a case where the number of workers present in the vicinity of the vehicle **100** is equal to or more than the predetermined number. The deficiency determination unit **213** may determine whether the worker gets on the vehicle **100** by analyzing the image acquired from the camera, and predict that the deficiency KS will occur in the measured appearance information MG in a case where the manufacturing process is the manufacturing process in which the worker gets on the vehicle **100** to perform the work.

Determination Method A4

[0072] In the determination method A4, the deficiency determination unit **213** determines that the deficiency KS has occurred in the measured appearance information MG in a case where the deficiency KS that has occurred in the measured appearance information MG is detected. The deficiency determination unit **213** detects that the deficiency KS has occurred in the measured appearance information MG by using measured appearance information MG1 used for current matching, measured appearance information MG2 used for matching N times ago (N is a natural number), and the movement information MV stored in the memory **202**. Here, the measured appearance information MG1 used for the current matching is the measured appearance information MG measured by the external sensor **300** at time T1, and the measured appearance information MG2 used for the matching N times ago is the measured appearance information MG measured by the external sensor **300** at time T2 earlier than time T1. As shown in FIG. 7, the movement information MV includes time-series data of the speed and the steering angle of the vehicle **100** measured by the internal sensor group **140** from time T2 to time T1. As shown in FIG. 8, first, the deficiency determination unit **213** decides a bounding box BB surrounding a portion,

corresponding to the vehicle **100**, in the measured appearance information MG2 used for the matching N times ago. A portion, corresponding to the vehicle **100**, in the measured appearance information MG2 is a portion that matches with the reference appearance information RG in the matching. Next, the deficiency determination unit **213** calculates a movement amount and a rotation amount of the vehicle **100** between time T2 and time T1 by using the movement information MV stored in the memory **202**. The deficiency determination unit **213** moves the bounding box BB by the movement amount of the vehicle **100** from time T2 to time T1, and rotates the bounding box BB by the rotation amount of the vehicle **100** from time T2 to time T1. The deficiency determination unit **213** acquires the number of point clouds of the portion surrounded by the bounding box BB in the measured appearance information MG1 used for the current matching. The number of point clouds in the portion surrounded by the bounding box BB is smaller in a case where the deficiency KS has occurred in the measured appearance information MG1 than in a case where the deficiency KS has not occurred. Therefore, the deficiency determination unit **213** detects that the deficiency KS has occurred in the measured appearance information MG1 in a case where the number of point clouds in the portion surrounded by the bounding box BB is equal to or less than a predetermined number.

[0073] In the determination method A4, the deficiency determination unit **213** can calculate the movement amount of the vehicle **100** by Expression (1). In Expression (1), x represents the movement amount of the vehicle **100** in an X direction, y represents the movement amount of the vehicle **100** in a Y direction, t represents time, v represents the speed of the vehicle **100**, and θ represents the steering angle of the vehicle **100**. Expression (1) represents the movement amount of the vehicle **100** in two seconds.

$$[00001] \ (x(t), y(t)) = (x(t-2) + \int_{t-2}^t v(t) \cos \theta dt, y(t-2) + \int_{t-2}^t v(t) \sin \theta dt) \quad (1)$$

[0074] As shown in FIG. 5, in a case where a determination is made in step S120 that the deficiency KS has occurred in the measured appearance information MG, the complement unit **214** complements the deficiency in the measured appearance information MG in step S130. In the present embodiment, the complement unit **214** complements the deficiency in the measured appearance information MG1 used for the current matching by using the measured appearance information MG2 used for the matching N times ago and the movement information MV stored in the memory **202**. Here, the measured appearance information MG1 used for the current matching is the measured appearance information MG measured by the external sensor **300** at time T1, and the measured appearance information MG2 used for the matching N times ago is the measured appearance information MG measured by the external sensor **300** at time T2 earlier than time T1. The number of N is preferably smaller. The number of N is preferably 1. As shown in FIG. 7, the movement information MV includes the time-series data of the speed and the steering angle of the vehicle **100** measured by the internal sensor group **140** from time T2 to time T1. As shown in FIG. 9, first, the complement unit **214** calculates the movement amount and the rotation amount of the vehicle **100** between time T2 and time T1 by using the movement information MV stored in the memory **202**. Next, the complement unit **214** moves the measured appearance information MG2 by the movement amount of the vehicle **100** from time T2 to time T1, and rotates the measured appearance information MG2 by the rotation amount of the vehicle **100** from time T2 to time T1. By moving and rotating the measured appearance information MG2, a portion, corresponding to the vehicle **100**, in the measured appearance information MG2 overlaps a deficient portion in the measured appearance information MG1, and the deficiency KS in the measured appearance information MG1 is complemented. The complement unit **214** may move and rotate the entire measured appearance information MG2, may move and rotate solely a portion, corresponding to the vehicle **100**, in the measured appearance information MG2, or may move and rotate solely a portion, corresponding to the deficient portion in the measured appearance information MG1, in the measured appearance information MG2. The complement unit **214** may move the measured

appearance information MG2 without rotating the measured appearance information MG2 when the complement unit **214** complements the deficiency KS in the measured appearance information MG1.

[0075] In step S140, the estimation unit **215** estimates the position and the orientation of the vehicle **100** by executing the matching using the measured appearance information MG and the reference appearance information RG. When the complement of the deficiency KS is not executed in step S130, the estimation unit **215** estimates the position and the orientation of the vehicle **100** by executing the matching between the measured appearance information MG in which the complement of the deficiency KS is not executed with the reference appearance information RG. When the complement of the deficiency KS is executed in step S130, the estimation unit **215** estimates the position and the orientation of the vehicle **100** by executing the matching between the measured appearance information MG in which the complement of the deficiency KS is executed with the reference appearance information RG. As a method of the matching, for example, a normal distributions transform (NDT) or an iterative closest point (ICP) can be used. The estimation unit **215** can estimate the position and the orientation of the vehicle **100** in a coordinate system of the external sensor **300** by executing the matching. A position and an orientation of the external sensor **300** are fixed, and a positional relationship between the coordinate system of the external sensor **300** and the global coordinate system GC is known. Therefore, the estimation unit **215** can convert the position and the orientation of the vehicle **100** in the coordinate system of the external sensor **300** into the position and the orientation of the vehicle **100** in the global coordinate system GC. A process of acquiring the measured appearance information MG may be referred to as an appearance information acquisition process. A process of acquiring the movement information MV may be referred to as a movement information acquisition process. A process of determining whether the deficiency has occurred in the measured appearance information MG may be referred to as a determination process. A process of complementing the deficiency KS in the measured appearance information MG may be referred to as a complement process. A process of estimating at least one of the position and the orientation of the vehicle **100** may be referred to as an estimation process.

[0076] In step S150, the remote controller **216** decides a target position to which the vehicle **100** should head next. In the present embodiment, the target position is represented by the coordinates of X, Y, and Z in the global coordinate system GC. A reference route RR that is a route along which the vehicle **100** should travel is stored in advance in the memory **202** of the server device **200**. The route is represented by a node indicating a departure point, a node indicating a passing point, a node indicating a destination, and a link connecting the respective nodes. The remote controller **216** decides the target position to which the vehicle **100** should head next by using the vehicle position information and the reference route RR. The remote controller **216** decides the target position on the reference route RR ahead of a current position of the vehicle **100**.

[0077] In step S160, the remote controller **216** generates the traveling control signal for causing the vehicle **100** to travel toward the decided target position. The remote controller **216** calculates a traveling speed of the vehicle **100** from the transition of the position of the vehicle **100**, and compares the calculated traveling speed with a target speed. As a whole, the remote controller **216** decides the acceleration such that the vehicle **100** is accelerated when the traveling speed is lower than the target speed, and decides the acceleration such that the vehicle **100** is decelerated when the traveling speed is higher than the target speed. The remote controller **216** decides the steering angle and the acceleration such that the vehicle **100** does not deviate from the reference route RR when the vehicle **100** is located on the reference route RR, and decides the steering angle and the acceleration such that the vehicle **100** returns to the reference route RR when the vehicle **100** is not located on the reference route RR, in other words, when the vehicle **100** deviates from the reference route RR.

[0078] In step S170, the remote controller **216** transmits the generated traveling control signal to the vehicle **100**. The server device **200** repeatedly executes, at a predetermined cycle, the

acquisition of the vehicle position information, the decision of the target position, the generation of the traveling control signal, and the transmission of the traveling control signal.

[0079] In step **S180**, the traveling controller **196** receives the traveling control signal transmitted from the server device **200**. In step **S190**, the traveling controller **196** controls the actuator group **120** by using the received traveling control signal to cause the vehicle **100** travel at the acceleration and the steering angle represented by the traveling control signal. The vehicle control device **110** repeatedly executes, at a predetermined cycle, the reception of the traveling control signal and the control of the actuator group **120**.

[0080] The movement amount and the rotation amount of the vehicle **100**, which are calculated from the movement information MV as the measurement result of the internal sensor group **140** mounted in the vehicle **100**, may deviate from an actual movement amount and an actual rotation amount of the vehicle **100**. Therefore, in a case where the measured appearance information MG2 used for the matching N times ago is modified (the bounding box BB is moved and rotated) by the movement amount and the rotation amount calculated from the movement information MV to complement the deficiency KS in the measured appearance information MG1 used for the current matching, a deviation may occur between the measured appearance information MG1 and the measured appearance information MG2. Therefore, in a case where the deficiency KS is complemented in the measured appearance information MG2 used to complement the deficiency KS in the measured appearance information MG1, there is a possibility that the estimation accuracy of the position and the orientation of the vehicle **100** is decreased. In this way, when the complement of the measured appearance information MG using the complemented measured appearance information MG is repeatedly executed, there is a possibility that the amount of decrease in the estimation accuracy of the position and the orientation of the vehicle **100** is increased. Therefore, in the present embodiment, the remote controller **216** decelerates or stops the vehicle **100** in a case where the complement of the deficiency KS in the measured appearance information MG is repeatedly executed a predetermined number of times. The predetermined number of times may be fixed in advance or may be optionally decided (changed) by the remote controller **216**. The remote controller **216** can decide the predetermined number of times by using, for example, at least one of the following number-of-times decision methods B1 to B3.

Number-of-Times Decision Method B1

[0081] In the number-of-times decision method B1, the remote controller **216** decides the number of times in accordance with a movement state of the vehicle **100**. It is not preferable to move the vehicle **100** at a high speed in a state where the estimation accuracy of the position and the orientation of the vehicle **100** is decreased, since there is a high possibility that the vehicle **100** comes into contact with the obstacle or the like. Therefore, the remote controller **216** may reduce the number of times as the speed of the vehicle **100** is increased. In this case, it is possible to reduce the possibility that the vehicle **100** comes into contact with the obstacle or the like. In addition, in a case where the vehicle **100** is making a turn, the behavior of the vehicle **100** is likely to be unstable. Therefore, the remote controller **216** may reduce the predetermined number of times in a case where the vehicle **100** is making a turn than in a case where the vehicle **100** is traveling straight. In this case, a situation can be suppressed in which the behavior of the vehicle **100** is unstable.

Number-of-Times Decision Method B2

[0082] In the number-of-times decision method B2, the remote controller **216** decides the predetermined number of times in accordance with the magnitude of the deficiency KS in the measured appearance information MG. The estimation accuracy of the position and the orientation of the vehicle **100** is likely to be lower as the deficiency KS in the measured appearance information MG is larger. Therefore, the remote controller **216** may reduce the number of times as the deficiency KS in the measured appearance information MG is increased. In this case, a significant decrease in the estimation accuracy of the position and the orientation of the vehicle **100** can be suppressed.

Number-of-Times Decision Method B3

[0083] In the number-of-times decision method B3, the remote controller **216** decides the predetermined number of times in accordance with a distance between the external sensor **300** and the vehicle **100**. The estimation accuracy of the position and the orientation of the vehicle **100** is likely to be lower as the distance between the external sensor **300** and the vehicle **100** is larger. Therefore, the remote controller **216** may reduce the predetermined number of times as the distance between the external sensor **300** and the vehicle **100** is increased. In this case, a significant decrease in the estimation accuracy of the position and the orientation of the vehicle **100** can be suppressed.

[0084] With the system **10** according to the present embodiment described above, even when the deficiency KS has occurred in the measured appearance information MG acquired from the external sensor **300**, the deficiency KS in the measured appearance information MG can be complemented. Therefore, a situation can be suppressed in which a correct estimation result of the position or the orientation of the vehicle **100** cannot be obtained due to the occurrence of the deficiency KS in the measured appearance information MG.

B. Second Embodiment

[0085] FIG. **10** is a diagram showing a configuration of a system **10b** according to a second embodiment. FIG. **11** is a diagram showing a configuration of the vehicle **100** according to the second embodiment. As shown in FIG. **10**, the second embodiment is different from the first embodiment in that the system **10b** does not include the server device **200** and that the vehicle **100** is configured to be travel via the autonomous control of the vehicle **100** instead of the remote control. Other configurations are the same as the configurations of the first embodiment unless otherwise described. In the present embodiment, the vehicle control device **110** can be regarded as a “device” according to the present disclosure, and the traveling controller **196** can be regarded as a “controller” according to the present disclosure.

[0086] As shown in FIG. **11**, in the present embodiment, the communication device **130** communicates with the external sensor **300** and the process management device **400** via wireless communication. The reference route RR and the reference appearance information RG are stored in advance in the memory **112** of the vehicle control device **110**. In the present embodiment, the processor **111** of the vehicle control device **110** functions as a measured appearance information acquisition unit **191**, a movement information acquisition unit **192**, a deficiency determination unit **193**, a complement unit **194**, an estimation unit **195**, and a traveling controller **196** by executing the computer program PG1 stored in advance in the memory **112**.

[0087] The measured appearance information acquisition unit **191** acquires the measured appearance information MG from the external sensor **300**. The movement information acquisition unit **192** acquires the measurement result of the internal sensor group **140**, in other words, the movement information MV, and records the movement information MV in the memory **112**. The deficiency determination unit **193** determines whether the deficiency has occurred in the measured appearance information MG acquired by the measured appearance information acquisition unit **191**. The complement unit **194** uses the movement information MV to complement the deficiency KS in the measured appearance information MG in a case where a determination is made that the deficiency KS has occurred in the measured appearance information MG. The estimation unit **195** estimates at least one of the position and the orientation of the vehicle **100** by comparing the measured appearance information MG with the reference appearance information RG. The estimation unit **195** estimates both the position and the orientation of the vehicle **100**. In a case where the deficiency determination unit **193** does not determine that the deficiency KS has occurred in the measured appearance information MG, the estimation unit **195** compares the measured appearance information MG in which the deficiency KS is not complemented by the complement unit **194** with the reference appearance information RG. In a case where the deficiency determination unit **193** determines that the deficiency KS has occurred in the measured appearance information MG, the estimation unit **195** compares the measured appearance information MG in

which the deficiency KS is complemented by the complement unit **194** with the reference appearance information RG. In the present embodiment, the traveling controller **196** generates the traveling control signal for causing the vehicle **100** to travel by using the estimation result of the estimation unit **195**. The traveling controller **196** controls the actuator group **120** by using the traveling control signal generated by the traveling controller **196**.

[0088] FIG. **12** is a flowchart showing a processing procedure of traveling control of the vehicle **100** according to the second embodiment. Steps **S210** to **S270** are repeatedly executed by the processor **111** of the vehicle control device **110**. In step **S210**, the measured appearance information acquisition unit **191** acquires the measured appearance information MG from the external sensor **300**.

[0089] In step **S220**, the deficiency determination unit **213** determines whether the deficiency KS has occurred in the measured appearance information MG. The deficiency determination unit **193** determines whether the deficiency KS has occurred in the measured appearance information MG by using at least one of the determination methods **A1** to **A4**.

[0090] When a determination is made in step **S220** that the deficiency KS has occurred in the measured appearance information MG, the deficiency determination unit **193** advances the processing to step **S230**. When a determination is not made in step **S220** that the deficiency KS has occurred in the measured appearance information MG, the deficiency determination unit **193** skips step **S230** and advances the processing to step **S240**.

[0091] In step **S230**, the complement unit **194** complements the deficiency in the measured appearance information MG. In the present embodiment, the complement unit **194** complements the deficiency in the measured appearance information MG used for the current matching by using the measured appearance information MG used for the matching N times ago and the movement information MV stored in the memory **112**.

[0092] In step **S240**, the estimation unit **195** estimates the position and the orientation of the vehicle **100** by executing the matching using the measured appearance information MG and the reference appearance information RG. When the complement of the deficiency KS is not executed in step **S230**, the estimation unit **195** estimates the position and the orientation of the vehicle **100** by executing the matching between the measured appearance information MG in which the complement of the deficiency KS is not executed with the reference appearance information RG. When the complement of the deficiency KS is executed in step **S230**, the estimation unit **195** estimates the position and the orientation of the vehicle **100** by executing the matching between the measured appearance information MG in which the complement of the deficiency KS is executed with the reference appearance information RG.

[0093] In step **S250**, the traveling controller **196** decides the target position to which the vehicle **100** should head next. In step **S260**, the traveling controller **196** generates the traveling control signal for causing the vehicle **100** to travel toward the decided target position. In step **S270**, the traveling controller **196** controls the actuator group **120** by using the generated traveling control signal to cause the vehicle **100** to travel in accordance with the parameter represented by the traveling control signal. The processor **111** of the vehicle control device **110** repeatedly executes the acquisition of the vehicle position information, the decision of the target position, the generation of the traveling control signal, and the control of the actuator group **120** at a predetermined cycle.

[0094] With the system **10b** according to the present embodiment described above, the vehicle **100** can be caused to travel via the autonomous control of the vehicle **100** without the need for the server device **200** to remotely control the vehicle **100**. Further, in the present embodiment, as in the first embodiment, even when the deficiency KS has occurred in the measured appearance information MG acquired from the external sensor **300**, the deficiency KS in the measured appearance information MG can be complemented. Therefore, a situation can be suppressed in which a correct estimation result of the position or the orientation of the vehicle **100** cannot be

obtained due to the occurrence of the deficiency KS in the measured appearance information MG.

C. Other Embodiments

[0095] (C1) In the first and second embodiments, the external sensor **300** is the LiDAR. In contrast, the external sensor **300** may be the camera. In this case, the estimation units **215**, **195** may estimate the position and the orientation of the vehicle **100** by image matching using the image of the vehicle **100** output from the camera, instead of the point cloud matching using the three-dimensional point cloud data of the vehicle **100** output from the LiDAR.

[0096] (C2) In the first and second embodiments, the server device **200** executes the processing from the acquisition of the vehicle position information of the vehicle **100** to the generation of the traveling control signal. On the other hand, the vehicle **100** may execute at least a part of the processing from the acquisition of the vehicle position information of the vehicle **100** to the generation of the traveling control signal. For example, the following forms (1) to (3) may be adopted.

[0097] (1) The server device **200** may acquire the vehicle position information of the vehicle **100**, decide the target position to which the vehicle **100** should head next, and generate the route from the current position of the vehicle **100** represented by the acquired vehicle position information to the target position. The server device **200** may generate a route to the target position between the current position and the destination or may generate a route to the destination. The server device **200** may transmit the generated route to the vehicle **100**. The vehicle **100** may generate the traveling control signal for causing the vehicle **100** to travel on the route received from the server device **200**, and control the actuator group **120** by using the generated traveling control signal.

[0098] (2) The server device **200** may acquire the vehicle position information of the vehicle **100** and transmit the acquired vehicle position information to the vehicle **100**. The vehicle **100** may decide the target position to which the vehicle **100** should head next, generate the route from the current position of the vehicle **100** represented by the received vehicle position information to the target position, generate the traveling control signal such that the vehicle **100** travels on the generated route, and control the actuator group **120** by using the generated traveling control signal. In each of the above-described embodiments, the vehicle operation information may be the route from the current position of the vehicle **100** to the target position.

[0099] (3) In the above-described forms (1) and (2), an internal sensor may be mounted in the vehicle **100**, and a detection result output from the internal sensor may be used for at least one of the generation of the route and the generation of the traveling control signal. The internal sensor may include, for example, a camera, a LiDAR, a millimeter wave radar, an ultrasound sensor, a GPS sensor, an acceleration sensor, and a gyro sensor. For example, in the above-described form (1), the server device **200** may acquire the detection result of the internal sensor and reflect the detection result of the internal sensor in the route when the route is generated. In the above-described form (1), the vehicle **100** may acquire the detection result of the internal sensor and reflect the detection result of the internal sensor to the traveling control signal when the traveling control signal is generated. In the above-described form (2), the vehicle **100** may acquire the detection result of the internal sensor and reflect the detection result of the internal sensor to the route when the route is generated. In the above-described form (2), the vehicle **100** may acquire the detection result of the internal sensor and reflect the detection result of the internal sensor to the traveling control signal when the traveling control signal is generated.

[0100] (C3) In the first and second embodiments, an internal sensor may be mounted in the vehicle **100**, and a detection result output from the internal sensor may be used for at least one of the generation of the route and the generation of the traveling control signal. For example, the vehicle **100** may acquire the detection result of the internal sensor and reflect the detection result of the internal sensor to the route when the route is generated. The vehicle **100** may acquire the detection result of the internal sensor and reflect the detection result of the internal sensor to the traveling control signal when the traveling control signal is generated.

[0101] (C4) In the second embodiment, the vehicle **100** acquires the vehicle position information by using the detection result of the external sensor **300**. On the other hand, the internal sensor may be mounted in the vehicle **100**, and the vehicle **100** may acquire the vehicle position information by using the detection result of the internal sensor, decide the target position to which the vehicle **100** should head next, generate the route from the current position of the vehicle **100** represented by the acquired vehicle position information to the target position, generate the traveling control signal for traveling on the generated route, and control the actuator group **120** by using the generated traveling control signal. In this case, the vehicle **100** can travel without using the detection result of the external sensor **300** at all. The vehicle **100** may acquire a target arrival time or traffic jam information from the outside of the vehicle **100** and reflect the target arrival time or the traffic jam information to at least one of the route and the traveling control signal.

[0102] (C5) In the first and second embodiments, the server device **200** automatically generates the traveling control signal to be transmitted to the vehicle **100**. On the other hand, the server device **200** may generate the traveling control signal to be transmitted to the vehicle **100** in response to an operation of an external operator who is located outside the vehicle **100**. For example, the external operator may operate an operation device including a display that displays the captured image output from the camera as the external sensor **300**, a steering wheel, an accelerator pedal, and a brake pedal for remotely operating the vehicle **100**, and a communication device that communicates with the server device **200** via wired communication or wireless communication, and the server device **200** may generate the traveling control signal in response to the operation applied to the operation device.

[0103] (C6) In the first and second embodiments, the vehicle **100** need solely have a configuration capable of moving via the unmanned driving, and may have, for example, a form of a platform having a configuration described below. Specifically, the vehicle **100** need solely include at least the vehicle control device **110** and the actuator group **120**, in order to exhibit the three functions of “traveling”, “turning”, and “stopping” via the unmanned driving. In a case where the vehicle **100** acquires the information from the outside for the unmanned driving, the vehicle **100** need solely further include the communication device **130**. That is, the vehicle **100** that can move via the unmanned driving need not be mounted with at least a part of the interior components, such as the driver's seat and the dashboard, need not be mounted with at least a part of the exterior components, such as the bumper and the fender mirror, and need not be mounted with the bodyshell. In this case, the remaining components, such as the bodyshell, may be mounted in the vehicle **100** before the vehicle **100** is shipped from a factory FC, or the remaining components, such as the bodyshell, may be mounted in the vehicle **100** after the vehicle **100** is shipped from the factory FC in a state where the remaining components, such as the bodyshell, are not mounted in the vehicle **100**. Each component may be mounted from any direction, such as the upper side, the lower side, the front side, the rear side, the right side, or the left side of the vehicle **100**, and may be mounted from the same direction or different directions. The position decision can be made for the form of the platform in the same manner as the vehicle **100** according to the first embodiment.

[0104] (C7) The vehicle **100** may be manufactured by combining a plurality of modules. The module means a unit configured by a plurality of components assembled depending on the part or the function of the vehicle **100**. For example, the platform of the vehicle **100** may be manufactured by combining a front module that constitutes a front portion of the platform, a center module that constitutes a center portion of the platform, and a rear module that constitutes a rear portion of the platform. In addition, the number of modules constituting the platform is not limited to three, and may be two or less or four or more. In addition to or instead of the components constituting the platform, the components constituting a portion of the vehicle **100** that is different from the platform may be modularized. In addition, various modules may include any exterior component, such as a bumper or a grille, or any interior component, such as a seat or a console. The present disclosure is not limited to the vehicle **100**, a mobile body of any aspect may be manufactured by

combining the modules. Such a module may be manufactured, for example, by joining the components via welding, a fastener, or the like, or may be manufactured by integrally molding at least a part of the components constituting the modules as one component via casting. A molding method of integrally molding one component, particularly a relatively large component, is also called giga casting or mega casting. For example, the front module, the center module, and the rear module may be manufactured by using giga casting.

[0105] (C8) The transport of the vehicle **100** using the traveling of the vehicle **100** via the unmanned driving is also referred to as “autonomous transport”. A configuration for implementing the autonomous transport is also referred to as “vehicle remote control autonomous driving transport system”. A production method of producing the vehicle **100** by using the autonomous transport is also referred to as “autonomous production”. In the autonomous production, for example, at the factory FC that manufactures the vehicle **100**, at least a part of the transport of the vehicle **100** is implemented by the autonomous transport.

[0106] (C9) In the first and second embodiments, a part or all of the functions and the processing that are implemented by software may be implemented by hardware. Alternatively, a part or all of the functions and the processing that are implemented by hardware may be implemented by software. As the hardware for implementing various functions in each of the above-described embodiments, for example, various circuits, such as an integrated circuit or a discrete circuit, may be used.

[0107] The present disclosure is not limited to the above-described embodiments, and can be implemented with various configurations without departing from the gist of the present disclosure. For example, the technical features in the embodiments corresponding to the technical features in the respective forms described in SUMMARY can be replaced or combined as appropriate to solve a part or all of the above-described objects, or to achieve a part or all of the above-described effects. In a case where the technical features are not described as being always needed in the present specification, the features can be deleted as appropriate.

Claims

1. A device comprising: an appearance information acquisition unit configured to acquire first measured appearance information on an appearance of a mobile body from a detector configured to measure the appearance of the mobile body; a movement information acquisition unit configured to acquire movement information on movement of the mobile body; a determination unit configured to determine whether a deficiency has occurred in the first measured appearance information; a complement unit configured to, in a case where the determination unit determines that the deficiency has occurred, use the movement information to complement the deficiency in the first measured appearance information; and an estimation unit configured to estimate at least one of a position and an orientation of the mobile body by comparing reference appearance information on the appearance of the mobile body with the first measured appearance information, the estimation unit being configured to, in the case where the determination unit determines that the deficiency has occurred, compare the first measured appearance information complemented by the complement unit with the reference appearance information.
2. The device according to claim 1, wherein the determination unit is configured to, in a case where the deficiency is detected from the first measured appearance information, determine that the deficiency has occurred in the first measured appearance information.
3. The device according to claim 1, wherein the determination unit is configured to, in a case where the determination unit predicts that the deficiency will occur in the first measured appearance information, determine that the deficiency has occurred in the first measured appearance information.
4. The device according to claim 3, wherein the determination unit is configured to, in at least one

of a case where the mobile body is located at a predetermined place and a case where the mobile body passes through the predetermined place, predict that the deficiency will occur in the first measured appearance information.

5. The device according to claim 4, wherein the predetermined place is a place where a predetermined number of workers are present.

6. The device according to claim 4, wherein the predetermined place is a place where a worker gets on the mobile body to perform work.

7. The device according to claim 1, further comprising a controller configured to, in a case where the complement of the deficiency in the first measured appearance information is repeated a predetermined number of times, decelerate or stop the mobile body.

8. The device according to claim 7, wherein the controller is configured to change the predetermined number of times in accordance with a movement state of the mobile body.

9. The device according to claim 8, wherein the controller is configured to execute at least one of (A) reducing the predetermined number of times as a speed of the mobile body is increased, and (B) reducing the predetermined number of times in a case where the mobile body is making a turn than in a case where the mobile body is traveling straight.

10. The device according to claim 7, wherein the controller is configured to change the predetermined number of times in accordance with magnitude of the deficiency.

11. The device according to claim 10, wherein the controller is configured to reduce the predetermined number of times as the magnitude of the deficiency is increased.

12. The device according to claim 7, wherein the controller is configured to change the predetermined number of times in accordance with a distance between the detector and the mobile body.

13. The device according to claim 12, wherein the controller is configured to reduce the predetermined number of times as the distance between the detector and the mobile body is increased.

14. The device according to claim 1, wherein the first measured appearance information is three-dimensional point cloud data measured by the detector within a predetermined period.

15. The device according to claim 1, further comprising a storage medium configured to store the reference appearance information, wherein the reference appearance information is three-dimensional point cloud data generated by using CAD data representing the appearance of the mobile body.

16. The device according to claim 1, further comprising a storage medium configured to store the movement information, wherein the complement unit is configured to complement the deficiency in the first measured appearance information used for current matching by using second measured appearance information used for matching N times ago and the movement information.

17. The device according to claim 1, wherein the complement unit is configured to complement the deficiency in the first measured appearance information by modifying information on a movement amount of the mobile body in second measured appearance information by an amount corresponding to a movement amount of the mobile body within a predetermined period, and modifying information on a rotation amount of the mobile body in the second measured appearance information by an amount corresponding to a rotation amount of the mobile body within the predetermined period.

18. The device according to claim 1, wherein: the estimation unit is configured to, in a case where the complement unit does not complement the deficiency, estimate the position and the orientation of the mobile body by executing matching between the first measured appearance information in which the deficiency is not complemented and the reference appearance information; and the estimation unit is configured to, in the case where the complement unit complements the deficiency, estimate the position and the orientation of the mobile body by comparing the first measured appearance information in which the deficiency is complemented, with the reference

appearance information.

19. A system comprising: a detector configured to measure an appearance of a mobile body; an appearance information acquisition unit configured to acquire measured appearance information on the appearance of the mobile body from the detector; a movement information acquisition unit configured to acquire movement information on movement of the mobile body; a determination unit configured to determine whether a deficiency has occurred in the measured appearance information; a complement unit configured to, in a case where the determination unit determines that the deficiency has occurred, use the movement information to complement the deficiency in the measured appearance information; and an estimation unit configured to estimate at least one of a position and an orientation of the mobile body by comparing reference appearance information on the appearance of the mobile body with the measured appearance information, the estimation unit being configured to, in the case where the determination unit determines that the deficiency has occurred, compare the measured appearance information complemented by the complement unit with the reference appearance information.

20. A method comprising: acquiring measured appearance information on an appearance of a mobile body by measuring the appearance of the mobile body; acquiring movement information on movement of the mobile body; determining whether a deficiency has occurred in the measured appearance information; complementing, in a case where a determination is made that the deficiency has occurred, the deficiency in the measured appearance information by using the movement information; and estimating, in the case where the determination is made that the deficiency has occurred, at least one of a position and an orientation of the mobile body by comparing the complemented measured appearance information with reference appearance information on the appearance of the mobile body.
